
Uni-variate (1D) inference about **two-sample** normal population means

PAIRED comparison

Model	Pairwise observations $(X_{1i}, X_{2i}), i = 1, 2, \dots, n$ Population 1: $X_{11}, X_{12}, \dots, X_{1n}$ iid $X_{1i} \sim N(\mu_1, \sigma_1^2)$ Population 2: $X_{21}, X_{22}, \dots, X_{2n}$ iid $X_{2i} \sim N(\mu_2, \sigma_2^2)$ $\mu_1, \sigma_1^2, \mu_2, \sigma_2^2$ unknown X_{1i}, X_{2i} not independent
Difference model	$D_i = X_{1i} - X_{2i}$ iid $D_i \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \stackrel{\text{def}}{=} N(\mu_1 - \mu_2, \sigma_D^2)$
Hypothesis	$H_0: \mu_1 - \mu_2 = \delta$ against $H_1: \mu_1 - \mu_2 \neq \delta$
Point-estimators	$\hat{\mu}_1 - \hat{\mu}_2 = \bar{D} = \frac{1}{n} \sum_{i=1}^n D_i \sim N(\mu_1 - \mu_2, \frac{\sigma_D^2}{n})$ $\hat{\sigma}_D^2 = S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2 \sim \frac{\sigma_D^2}{n-1} \chi_{n-1}^2$
Test-statistic	$H_0 \Rightarrow T = \frac{\bar{D} - \delta}{S_D / \sqrt{n}} \sim t_{n-1}$
Test of hypothesis	calculate value of test-statistic $T = t$ <u>approach I:</u> Select level of significance, α (level of confidence $1 - \alpha$) Reject H_0 if $ t > t_{n-1, \alpha/2}$ <u>approach II:</u> calculate p-value: $p = 2P(t_{n-1} > t)$ Reject H_0 for all $\alpha > p$
CI for $(\mu_1 - \mu_2)$	$P\left(-t_{n-1, \alpha/2} < \frac{\bar{D} - (\mu_1 - \mu_2)}{S_D / \sqrt{n}} < t_{n-1, \alpha/2}\right) = 1 - \alpha \Rightarrow$ 100(1 - α)% confidence interval for $(\mu_1 - \mu_2)$: $\left[\bar{D} - \frac{S_D}{\sqrt{n}} t_{n-1, \alpha/2}, \bar{D} + \frac{S_D}{\sqrt{n}} t_{n-1, \alpha/2}\right]$

Uni-variate (1D) inference about **two-sample** normal population means

NON-PAIRED comparison (two independent samples)

CASE 1: EQUAL POPULATION VARIANCES

Model **Population 1 obs:** $X_{11}, X_{12}, \dots, X_{1n_1}$ iid $X_{1i} \sim N(\mu_1, \sigma^2)$
Population 2 obs: $X_{21}, X_{22}, \dots, X_{2n_2}$ iid $X_{2i} \sim N(\mu_2, \sigma^2)$
 μ_1, μ_2, σ^2 unknown
 X_{1i}, X_{2j} **independent**

Hypothesis $H_0: \mu_1 - \mu_2 = \delta$ against $H_1: \mu_1 - \mu_2 \neq \delta$

Point-estimators $\hat{\mu}_1 = \bar{x}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_{1i} \sim N(\mu_1, \frac{\sigma^2}{n_1})$
 $\hat{\mu}_2 = \bar{x}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{2i} \sim N(\mu_2, \frac{\sigma^2}{n_2})$
Two estimates for $\hat{\sigma}^2$ available:
 $\hat{\sigma}^2 = S_1^2 = \frac{1}{n_1-1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2 \sim \frac{\sigma^2}{n_1-1} \chi_{n_1-1}^2$
 $\hat{\sigma}^2 = S_2^2 = \frac{1}{n_2-1} \sum_{i=1}^{n_2} (x_{2i} - \bar{x}_2)^2 \sim \frac{\sigma^2}{n_2-1} \chi_{n_2-1}^2$
Improved combined (**pooled**) estimate for $\hat{\sigma}^2$:
 $\hat{\sigma}^2 = S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \sim \frac{\sigma^2}{n_1+n_2-2} \chi_{n_1+n_2-2}^2$

Test-statistic $H_0 \Rightarrow T = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$

Test of hypothesis calculate value of test-statistic $T = t$
approach I:
Select level of significance, α (level of confidence $1 - \alpha$)
Reject H_0 if $|t| > t_{n_1+n_2-2, \alpha/2}$
approach II:
calculate p-value: $p = 2P(t_{n_1+n_2-2} > |t|)$
Reject H_0 for all $\alpha > p$

CI for $(\mu_1 - \mu_2)$ $P\left(-t_{n_1+n_2-2, \alpha/2} < \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} < t_{n_1+n_2-2, \alpha/2}\right) = 1 - \alpha \Rightarrow$
100(1 - α)% confidence interval for $(\mu_1 - \mu_2)$:
 $\left[(\bar{x}_1 - \bar{x}_2) - S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} t_{n_1+n_2-2, \alpha/2}, (\bar{x}_1 - \bar{x}_2) + S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} t_{n_1+n_2-2, \alpha/2} \right]$

TEST FOR EQUAL VARIANCES

Hypothesis	$H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$
Point-estimators	$S_1^2 = \frac{1}{n_1-1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2 \sim \frac{\sigma_1^2}{n_1-1} \chi_{n_1-1}^2$ $S_2^2 = \frac{1}{n_2-1} \sum_{i=1}^{n_2} (x_{2i} - \bar{x}_2)^2 \sim \frac{\sigma_2^2}{n_2-1} \chi_{n_2-1}^2$
Test-statistic	$H_0 \Rightarrow T = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$

CASE 2: UN-EQUAL POPULATION VARIANCES

Model	Population 1 obs: $X_{11}, X_{12}, \dots, X_{1n_1}$ iid $X_{1i} \sim N(\mu_1, \sigma_1^2)$ Population 2 obs: $X_{21}, X_{22}, \dots, X_{2n_2}$ iid $X_{2i} \sim N(\mu_2, \sigma_2^2)$ $\mu_1, \sigma_1^2, \mu_2, \sigma_2^2$ unknown X_{1i}, X_{2j} <u>independent</u>
Hypothesis	$H_0: \mu_1 - \mu_2 = \delta$ against $H_1: \mu_1 - \mu_2 \neq \delta$
Test-statistic	<u>No exact test available</u> If both samples are large, $n_1, n_2 \gg 1$, <u>approximate test</u> available $T = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0,1)$

WHY IS PAIRED COMPARISON BETTER THAN NON-PAIRED ?

Intuitive explanation	Paired comparison reduces unit-to-unit random variation
Mathematical explanation	Assume $n_1 = n_2 = n$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$ <u>Non-paired comparison:</u> $\text{Var}[\bar{x}_1 - \bar{x}_2] = 2 \frac{\sigma^2}{n}$ <u>Paired comparison:</u> $\text{Var}[\bar{x}_1 - \bar{x}_2] = 2 \frac{\sigma^2}{n} (1 - \rho_{12})$ $< 2 \frac{\sigma^2}{n}$ for $\rho_{12} > 0$
